# Confidence-Driven Region Mixing for Optical Remote Sensing Domain Adaptation Object Detection

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***Abstract*— Object detection is a challenging task that aims to locate and classify instances in an image simultaneously. High-performing deep detectors are typically trained on extensive labeled datasets and will face the problem of accuracy degradation in different scenarios. Through domain adaptation, the detector can attain generalization, making precise detections even in the presence of different data distributions without labels. In this article, we propose a confidence-driven region mixing (CR-Mixing) framework, which is a novel approach that integrates the region-selective mixing (RSM) method into a mean-teacher framework with an unbiased adversarial module (UAM), addressing the domain-shift and knowledge transfer problems in the unsupervised domain adaptation (UDA) object detection task. First, we construct RSM that utilizes pseudodetection labels to compute object information at the region level and integrates this information in a balanced manner to maximize accurate foreground details for the detection model. This enables the detection model to adjust its focus to unlabeled domains gradually. Second, to acquire a backbone network with object-focused features, we develop the UAM method. This method is a multiscale alignment network designed to constrain the backbone network in learning domain-invariant features. We conducted extensive experiments on two benchmarks, and the final results show that the CR-Mixing model demonstrates significant effectiveness in robust UDA for optical remote sensing object detection tasks.**

***Index Terms*— Knowledge transferring, object detection, unsupervised domain adaptation (UDA).**

## I. Introduction

REMOTE sensing object detection has been one of the most important tasks for decades, aiming to localize and classify objects in remote sensing images [1], [2], [3], [4]. In recent years, owing to the emergence and development of deep neural networks, remarkable progress has been achieved on large-scale benchmarks [5], [6], [7], [8]. Due to the versatility and superior accuracy of object detection, it is used for various tasks in remote sensing, such as traffic monitoring and flow management [9], [10], [11], natural disaster monitoring and response [12], [13], and military intelligence and security applications [14], [15], [16].

Nevertheless, prevailing object detection tasks in mainstream applications often rely on extensive datasets annotated with labels for training to achieve highly accurate results [17], [18], [19]. Some domain-constrained models may experience performance degradation in various sensors, geographic areas, atmospheric conditions, ground sampling distances, and sensor shooting angles [20]. Annotating the data is easier for common object images such as the MS-COCO dataset [21], but for remote sensing images where tens or even hundreds of objects exist in each image, fine annotation presents a significant challenge and demands considerable time and effort.

The development of algorithms that can transfer acquired knowledge from a labeled dataset (i.e., source domain) to an unlabeled dataset (i.e., target domain) is of growing significance within the realm of object detection. In response to this challenge, contemporary research efforts have focused on developing unsupervised domain adaptation (UDA) methods to reduce the gap between a source domain rich in labeled data and a target domain lacking labels, a phenomenon known as domain-shift [22], [23]. These efforts aim to address the domain-shift issue without relying on ground-truth labels in the target domain. Facilitating the transfer of acquired knowledge from the source domain to the target domain and aligning the distributions of both domains in an unsupervised manner empower the model to effectively generalize to target-domain data without the need for additional annotations [24], [25], [26]. To systematically and strategically confront the challenge of achieving accurate detection in the target domain, it is essential to reexamine the following two inquiries.

### A. Limited Labeled Data Versus Unlimited Remote Sensing Data

The limited availability of labeled data in the source domain, in contrast to the infinite unlabeled data in the unknown domain, poses a significant challenge. While achieving superior results might be more straightforward with the utilization of large-scale labeled data as a benchmark, this approach is not ideal. It tends to prioritize an abundance of labels rather than effectively reducing domain disparities in a trainable

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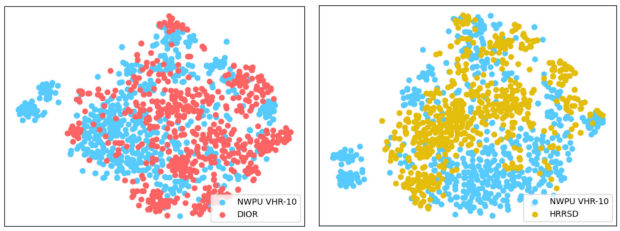
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(a) (b)

Fig. 1. (a) Distribution comparison between the NWPU VHR-10 dataset and the DIOR dataset. (b) Distribution comparison between the NWPU VHR-10 dataset and the HRRSD dataset. The distribution is calculated and displayed using 2-D t-SNE [27]. It can be observed that there are significant distributional differences between the datasets, making it challenging for the detection models to achieve satisfactory results.

manner and fully utilizing the knowledge of the target domain. Consequently, our goal is to achieve improved detection results with less dependence on labeled data by focusing on domain-invariant features.

### *B. Large Distribution Distance Between Source and Target Domains*

A detector trained for a specific scenario may fail to accurately detect objects in different scenarios due to significant variations in sample distribution under varying conditions, leading to performance degradation. Analyzing and solving the domain-shift problem are crucial for expanding the applicability of object detectors to a broader array of applications. By mapping data from various scenarios to the feature space and analyzing their distributions, it becomes evident that there is a gap between the distributions of data from different domains, as illustrated in Fig. 1.

To address these problems, we propose the confidence-driven region mixing (CR-Mixing) framework, a novel domain adaptation object detection framework for various types of optical remote sensing images. Instead of choosing to calculate distributional loss and directly aligning the source domain and the target domain [28], we build a deep mean-teacher framework based on the precise and efficient advanced model YOLOv5 [29] and enhanced it with a region-selective mixing (RSM) strategy and a multiscale unbiased adversarial module (UAM). By adaptively selecting regions from source- and target-domain images based on the confidence level to generate hybrid samples, the RSM strategy produces samples that contain the most relevant object information and establishes a connection between the two domains through visual similarity. It can enhance the generalization capability of the detector, as shown in Fig. 2. Meanwhile, to address the significant disparity in sample numbers between the two domains, we reuse source-domain samples mixed with various target-domain samples. This approach ensures that the labels are consistently constrained throughout the training process, thereby preventing the model from forgetting the accurate sample knowledge of the source domain. In response to the limitations of the mean-teacher framework in addressing the domain-shift problem, we formulate UAM that aims to assist backbone networks in learning domain-invariant features and reducing the distribution distance between two domains. Ultimately, we selected a

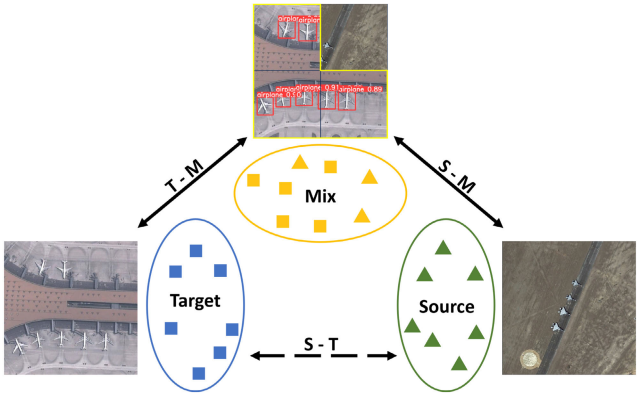



Fig. 2. Directly aligning the source domain and the target domain is challenging. However, by creating a mixed domain, the disparity between the two domains can be reduced, making it easier to transfer the acquired information from the source domain.

small sample size dataset as the source domain and two other expansive datasets as the target domains. Our goal is to achieve more robust experimental results with a reduced sample count.

The main contributions of this article are described as follows.

1) We implemented an RSM strategy within the mean-teacher framework to selectively mix images from the source domain and the target domain. It could not only fully take advantage of source-domain labels but also construct images that maximize foreground object information by mixing them based on the confidence of pseudolabels.
2) We construct a multiscale UAM during training to ensure that the model focuses on domain-invariant features. Considering the mixed domain encompassing various domain labels, we designed a pixel-level domain classifier in an unbiased manner to operate in conjunction with an RSM approach.
3) Our method achieves excellent performance results on two benchmarks for the task of optical remote sensing domain adaptation object detection. Ablation experiments further validate the effectiveness of the various modules employed in our proposed method.

The subsequent sections of this article are structured as follows. Section II provides an overview of related work. Section III elucidates the structure of the proposed method, along with the training methodology. Section IV includes comprehensive experimental results, visualizations, and ablation studies. Section V summarizes and analyzes the entire method.

## II. Related Works

Within this section, we provide a brief overview of pertinent methods in three dimensions: object detection, UDA, and domain adaptation object detection.

### *A. Object Detection*

With the development of deep neural networks, object detection tasks have demonstrated excellent performance and are being used in many applications. Nowadays, object detection methods can be grouped into two main categories:

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convolutional neural network (CNN)-based methods and transformer-based methods [30].

CNN-based object detectors, such as YOLO [7] and Faster R-CNN [6], are well-known methods. Their development and derivation continue to achieve the highest levels of detection accuracy today. CNN-based methods can also be divided into two main groups: one-stage detectors, such as YOLO [7] and SSD [31], use a unified framework to directly generate final results from feature maps. Two-stage object detectors, such as Faster R-CNN [6], generate predictions by initially identifying potential regions and then using this information to determine classification labels and bounding box coordinates.

Transformer-based methods are emerging, excellent feature extraction backbone neural networks. Given the remarkable success of transformer models in natural language processing (NLP) [30], their impressive capabilities have led to their adoption in computer vision and object detection. Vision Transformer (ViT) [32] utilizes transformers for image classification. The emergence of detection transformer (DETR) [8] has established a new paradigm for transformer-based object detection.

### B. Unsupervised Domain Adaptation

Given a source domain with labels and a target domain without labels, UDA aims to utilize the labeled data in the source domain to build a model that can generalize effectively and achieve high performance on the target domain. It was first widely explored in the field of image classification. The basic strategy involves measuring the distribution distance between different domains and training a feature extractor to minimize this distance. This primarily includes distance metric learning [33], adversarial training methods [28], and the mean-teacher model [34].

Some previous works aim to minimize the distance metric using various measurements, such as maximum mean discrepancy (MMD) [33], correlation alignment (CORAL), central moment discrepancy (CMD), Wasserstein distance, and transport distance. However, the primary purpose of these methods is for categorization tasks.

Adversarial training methods leverage generative adversarial networks (GANs) [35], which establish a confrontational scenario. In this setup, a discriminator determines whether an image is fake or real, while a generator generates images to deceive the discriminator. Ganin proposed a method that added gradient reversal layers (GRLs) into the backbone of the model to confuse the domain discriminator [28]. The aim is to obtain domain-invariant representations.

The mean-teacher [34] is a strategy that utilizes knowledge distillation to establish a teacher model and a student model with the same structure. It guides the student model toward domain independence by utilizing the exponential moving average (EMA).

### C. Domain Adaptation Object Detection

Although many models have shown impressive results in image classification tasks, there has been comparatively less research on domain adaptation object detection. Chen et al. [36] proposed DA-Faster [36] by exploring the untapped potential of conventional neural networks and has emerged as a trailblazer in cross-domain object detection tasks. DA-Faster improves the Faster R-CNN by incorporating domain-adaptive capabilities. It is the pioneering method that utilizes adversarial techniques to achieve domain adaptation object detection. Since then, the task has continued to develop and can be classified into three categories [37].

### D. Adversarial-Based Domain-Invariant Feature Learning

This adaptation approach involves adversarial training for the object detector model, facilitated by a domain discriminator and GRL proposed by Ganin and Lempitsky [28]. DA-Faster [36], SWDA [38], Mega-cda [39], HCL [40], HTCN [41], Every Pixel Matters [42], and RFA-Net [43] are all successors of adversarial-based domain-invariant feature learning methods following the DA-Faster.

### E. Mean-Teacher Training

Mean-teacher [34] is a powerful method for leveraging unlabeled data to enhance model generalization. It achieves this by iteratively training a detector model within a student–teacher framework. This has inspired several studies in the literature to investigate mean-teacher training. MTOR [44] introduced the mean-teacher framework to explore object relationships by focusing on region-level consistency, intergraph consistency, and intragraph consistency. UMT [45], AT [46], and HT [47] all achieved enhanced detection results by utilizing mean teacher models.

### F. Others

Alternative methods for domain adaptation object detection, such as pseudolabel self-training [48], [49], image-to-image translation [41], [50], [51], and domain randomization [51], [52], have demonstrated notable success. Pseudolabel self-training involves utilizing strongly confident predictions obtained from a source-trained detector model to further fine-tune it on the target domain. Image-to-image translation involves using a model that operates on unpaired images to transform target images into a representation that resembles the source. The domain randomization strategy generates various stylized iterations of the source-domain data to train the detector model. This approach helps reduce bias toward any particular style and improves the ability of the model to generalize to the target domain.

Within the field of remote sensing, the study conducted by Koga et al. [53] introduced a domain-adaptive vehicle detector, with the goal of improving performance generalization in the target domain. In this study, CORAL and adversarial-based domain adaptation techniques were applied to measure the distance between domains. FADA [54] integrated adversarial-based foreground alignment with prototype-based confusing feature alignment to enhance domain adaptation for domain adaptation remote sensing object detection tasks. It is a new strategy to create synthetic datasets using a 3-D engine that combines real-world aerial imagery with synthetic 3-D models. However, there is a lack of in-depth analysis of realistic scenarios. RFA-Net [43] utilized pseudolabel

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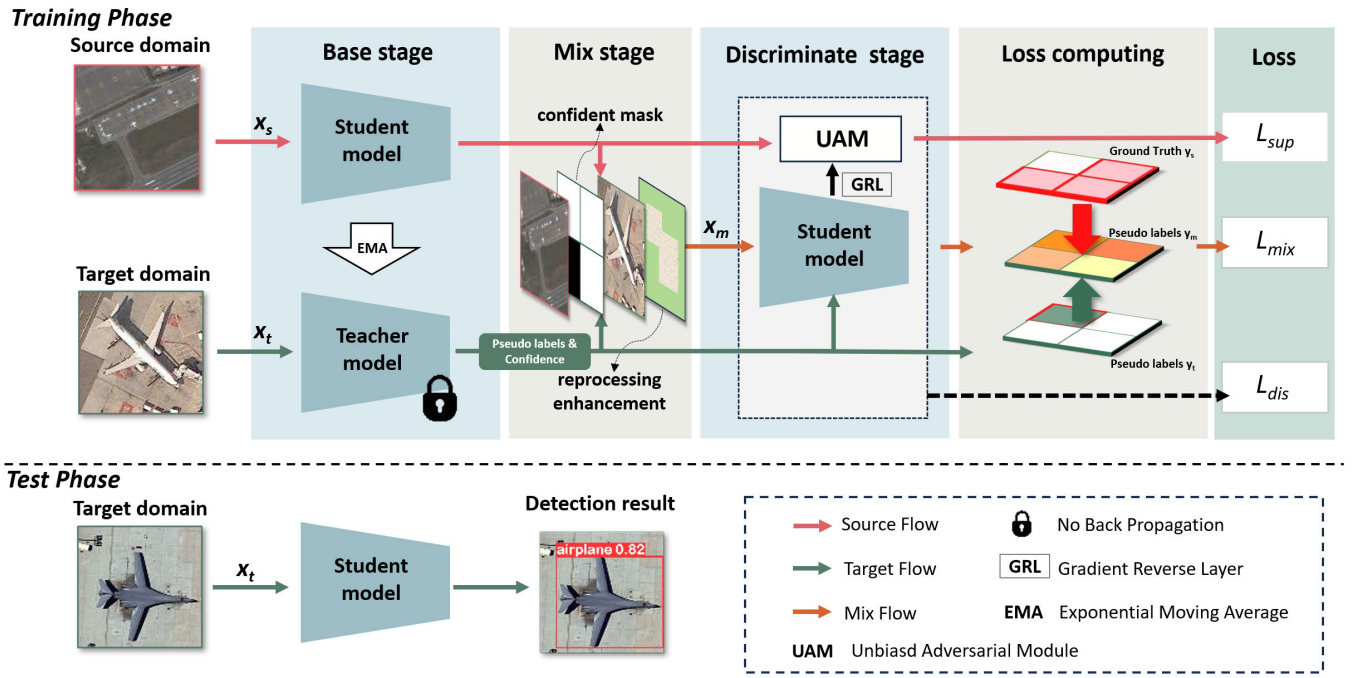



Fig. 3. Overall framework of our proposed method. In the training phase, we utilize the mean-teacher framework as the base model and integrate the RSM strategy and UAM into it. The RSM strategy balances the mix of the source domain and the target domain by utilizing the confidence output of the target-domain image passing through the teacher network as the basis. UAM is a multiscale domain classifier used to constrain the model to learn domain-invariant features. The confident mask is explained in Section III-C. The reprocessing enhancement represents a series of random augments. We also expand the source-domain data to maximize priori knowledge. During the testing phase, only the student model is used to predict the final outcome.

generation methods and adversarial-based training to facilitate optical remote sensing domain adaptation object detection tasks.

In the current field of domain adaptation object detection, the most commonly used baseline detection framework is still Faster R-CNN. However, even the most robust architecture of Faster R-CNN with the ResNet101 model as the backbone network falls short in efficiency and accuracy compared with more recent and streamlined models such as YOLOv5 in the realm of supervised object detection tasks. Thus, in this article, we utilize the efficient and precise one-stage object detector YOLOv5 as the baseline detection framework to develop the final CR-Mixing method.

## III. Methodology

The CR-Mixing framework is shown in Fig. 3 as an overview. It aims to construct a new mean-teacher structure using labeled source-domain data $D^s$ and unlabeled target-domain data $D^t$ with a confidence-based RSM strategy and a UAM to transfer domain-specific invariant knowledge from the source domain to the target domain and achieve multilevel domain feature alignment. We start from the mean-teacher model to discuss how to adapt it for domain adaptation object detection tasks. Then, we provide an analysis of the RSM strategy, and finally, we describe the UAM for domain-invariant feature extraction.

### A. Preliminary Definition

Domain adaptation object detection task involves the fully labeled source domain $D^s = \{X_s, Y_s\}$ and unlabeled target domain $D^t = \{X_t\}$. $Y_s$ represents the proper labels of source-domain images $X_s$, and the target domain only has images $X_t$ without labels. Generally, the distributions of these two domains have a big gap between them. Consequently, transferring directly without cross-domain alignment will result in poor generalization performance. The domain gap may arise due to variations in image sensors, capture time, weather conditions, platforms, illumination, and other factors [54].

In this work, we utilize the YOLOv5, an efficient object detection framework known for its higher accuracy, higher project integration, and faster speed compared to two-stage object detection models, as our baseline model. YOLOv5 incorporates three types of losses to calculate the total loss: $\mathcal{L}_{\text{box}}$ represents the complete-intersection over union (CIoU) loss for regressing the bounding box coordinates, $\mathcal{L}_{\text{obj}}$ denotes the binary cross-entropy (BCE) loss for the objectness score, and $\mathcal{L}_{\text{cl}}$ is a BCE loss for the classification score as

$$\mathcal{L}_{\text{sup}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{obj}}\mathcal{L}_{\text{obj}} + \lambda_{\text{cl}}\mathcal{L}_{\text{cl}} \tag{1}$$

where $\lambda_{\text{box}}$, $\lambda_{\text{obj}}$, and $\lambda_{\text{cl}}$ are the hyperparameters to balance the different weights of the three components.

### B. Mean-Teacher Framework

The mean-teacher framework [34] is a specific variant of knowledge distillation [55] that focuses on enhancing the training process of neural networks, particularly in the realm of semisupervised learning. Knowledge distillation is a broad concept used to transfer knowledge from complex models to simpler ones.

In the mean-teacher approach, the teacher model is identical to the student model, and a slight perturbation is introduced

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into the unlabeled data during the distillation process. By giving priority to samples with high prediction probabilities, the student model is motivated to reduce variance in the unlabeled target samples, enhancing the overall robustness of the model. This characteristic makes it particularly effective for object detection tasks. Simultaneously predicting detailed bounding boxes and object classes can make the model vulnerable to fluctuations in data. Furthermore, the student model undergoes updates through back-propagation, while the teacher model is updated using EMAs of the student's weights. Thus, the teacher network model can be seen as a historical average of the student network, which can improve training stability. The method can be formulated as follows:

$$\theta_t \leftarrow \alpha\theta_t + (1-\alpha)\theta_s \tag{2}$$

where $\alpha$ represents the exponential decay. Its reasonable value is close to 1.0, and in our method, we set it to 0.99. $\theta_t$ and $\theta_s$ represent the network parameters of the teacher model and the student model, respectively. The historical weights of the model can be well-preserved by a larger value of $\alpha$, thus maintaining the best stability among many historical model weight values.

The mean teacher model is a training approach similar to self-supervised contrastive learning, where a broader range of object representations are learned through contrasts. The traditional workflow of the mean-teacher model involves several steps. First, comparison samples are constructed from the target-domain data using various random augmentation techniques. Then, two different prediction samples are generated: one using the teacher network and the other using the student network. Finally, the model enforces domain independence by calculating the prediction differences to construct the consistency loss. This is to construct samples that behave differently but have the same ground labels. If the detector can effectively handle this pair of samples, it can be considered robust. However, there is a question that the mean-teacher model does not explicitly tackle the issue of domain-shift. The effect of random augmentation cannot be explicitly transferred from the source domain to the target domain. Consequently, the mean-teacher model is unable to directly acquire information about the target domain. Therefore, its utilization in domain adaptation object detection tasks may face challenges [46].

The substantial domain-shift could lead to unreliable predictions from the teacher model, thus reducing the effectiveness of the distillation process. Therefore, we designed the RSM method to select the region with the highest confidence in predicting the target-domain image and replace other places with labeled source-domain images. This ensures that the most accurate foreground object information is utilized, as explained in Section III-C. For the issue of the detector not focusing on the features of the target domain, we incorporate UAM to learn domain-invariant features, as elaborated on in Section III-D.

### C. Region-Selective Mixing Strategy

Based on the pseudodetections of the teacher model in the target domain, we designed a novel confidence-based RSM strategy to maximize the accuracy of foreground information content and enable multiple uses of source-domain data labels. Instead of randomly selecting a part of the target image [56] or choosing the region with the highest confidence [57] to mix, we first evaluate the foreground and background features in the four regions of each target-domain image $X_t$. We then design a more effective RSM strategy to adaptively mix the source- and target-domain images, as illustrated in Fig. 3 as the mix stage.

The decision to include a mix stage in the domain adaptation process serves as a strategic approach to create an auxiliary domain, enabling the mixing of data from two distinct domains into a unified one. This integration proves instrumental in aiding the migration of auxiliary model knowledge from the source domain to the target domain [57]. The specific calculation process and details can be seen in Algorithm 1.

**Algorithm 1** Region-Selective Mixing Through Adaptive Select and Reprocessing Enhancement

**Input:** $X_s \in$ Source domain, $X_t \in$ Target domain, setting thresholds $T_m, T_v$

**Output:** $X_m$ is the mixtures of Source domain and Target domain with enhancement

$L_{pseudo}, confidence =$ Model's prediction
Slices the target domain image into four regions
Calculate the mean values of the confidence of the predictions $Conf_i$ for each region
$Conf_m, Conf_v =$ The mean and variance value of $Conf_i$.
**if** $Conf_m \geq T_m$ and $Conf_v \leq T_v$ **then**
  $seq =$ **sort**$(Conf_i)$
  $\mathcal{M}^m =$ Build mix mask according to $seq$ that source domain $vs.$ target domain is 1:3
**else if** $Conf_m \geq T_m$ and $Conf_v > v_0$ **then**
  $seq =$ **sort**$(Conf_i)$
  $\mathcal{M}^m =$ Build mix mask according to $seq$ that source domain $vs.$ target domain is 1:1
**else**
  $seq =$ **sort**$(Conf_i)$
  $\mathcal{M}^m =$ Build mix mask according to $seq$ that source domain $vs.$ target domain is 3:1
**end if**
$X_m = \mathcal{M}^m \odot X_t + (1-\mathcal{M}^m) \odot X_s$
$X_m =$ **reprocessing enhancement**$(X_m)$
**return** $X_m$

To summarize, we first calculate the pseudolabel predictions using the teacher model, where each label includes location and confidence information. It is used as a basis for assessing the quantity and precision of targets in the region. We divide the entire target-domain image into four pieces, calculate the mean and variance of the confidence level separately, and develop three different mixing methods based on varying confidence scenarios. This ensures that the accurate foreground information of the target-domain images is maximally utilized, while the whole process is constrained by the source-domain images with labels. As for the reprocessing enhancement shown in Fig. 3, we randomly adjust brightness, contrast, saturation, and hue and apply random blurring to expand the representation of the mixed images. This exposes the model to a wider variety of possibilities, enhancing its robustness.

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In order to achieve optimal mixing results, it is essential to determine the two hyperparameters: $T_m$ and $T_v$.

The confidence-based mixing approach was selected because it recognizes that the information from the source domain may not show a strong adaptation in the target domain at the beginning of domain adaptation training. This means that relying on unknown and possibly incorrect answers to guide model updates is analogous to using prediction guidance. As a result, the poorly supervised confidence-based mixing strategy offers insightful information by focusing on particular regions without accurately finding objects. By taking the average of confidence levels, the model may determine the likelihood that a target falls into a particular category. This approach to weakly supervised training works well for guiding the model's overall growth in a positive direction. The particular mixing process can be expressed as follows:

$$X_m = \mathcal{M}^m \odot X_t + \left(1 - \mathcal{M}^m\right) \odot X_s \tag{3}$$

where $\odot$ is the dot product method, $X_t$, $X_s$, and $X_m$ are the target domain, the source domain, and mix images, and $\mathcal{M}^m$ is the confidence-based mask for the source- and target-domain images to mix. Building mixed images adaptively based on confidence level is a crucial part of the mix stage. One-quarter of the source domain and three-quarters of the target domain, one-half of the source domain and one-half of the target domain, or three-quarters of the source domain and one-quarter of the target domain are the three possible configurations for this mixed image. Confidence data computed from the target-domain data in the teacher model are used to determine the composition of the mixed image.

The predictions of the mixed images by the student model can be formulated as follows:

$$l_t^m, l_s^m = M_{\text{stu}}(X_m) \tag{4}$$

where $M_{\text{stu}}$ is the student model. $l_t^m$ and $l_s^m$ represent pseudo-detections of the different mixed regions of the source- and target-domain images. We construct a consistency loss on both the pseudodetections and the ground-truth labels as follows:

$$\mathcal{L}_{\text{mix}} = \mathcal{L}_{\text{box}}^t\left(l_t^m, l_t\right) + \mathcal{L}_{\text{box}}^s\left(l_s^m, Y_s^m\right) \tag{5}$$

where $\mathcal{L}_{\text{box}}^t(\cdot)$ and $\mathcal{L}_{\text{box}}^s(\cdot)$ represent the process of calculating the loss function for a subregion of the mix image. $\mathcal{L}_{\text{box}}^t(\cdot)$ is responsible for computing the target-domain region of the mixed image, comparing the predicted labels of the mixed image $l_t^m$ with the labels $l_t$ previously obtained through the teacher model to compute the loss function. $\mathcal{L}_{\text{box}}^t(\cdot)$ calculate the supervised loss by comparing the predicted labels of the mixed image $l_s^m$ with the source-domain ground-truth labels $Y_s$.

### D. Unbiased Adversarial Module

As labels are accessible for the source data, there is a propensity for both teachers and students to exhibit bias toward the source domain. Specifically, the pseudolabels generated by the teacher model for the target image are fundamentally derived from the knowledge acquired by the model trained with the source-domain labels. Consequently, it is imperative to mitigate the domain bias between the source and target domains. To address this issue, we incorporate adversarial learning into the mean-teacher framework to align the distributions of the two domains. The traditional methods are to add a domain classifier and CNN. To make the model learn more domain-invariant features and focus on the elements that the objects have, we propose a multiscale domain adaptive network for the backbone network CSPDarknet53 as the UAM to fully extract all the information about different scales of features. The detail can be seen in Fig. 4.

Instead of employing domain adaptation solely for the final scale of the feature extractor for abstractly aligning the two domains or for the early layer of the feature extractor for global alignment, as demonstrated in the DA-Faster architecture [36], we address the issue of gradient vanishing by extending domain adaptation to multiscales. In addition, if the single-scale domain adaptive framework is simply extended to a multiscale framework, the problem of inconsistent prediction results of domain classifiers at each scale may occur, thus affecting the domain adaptive performance. We propose a method that jointly combines multiscale domain features to ensure the consistency of the final results The detail can be seen in Algorithm 2.

**Algorithm 2** Calculate the $\mathcal{L}_{mix}$ by Unbiased Adversarial Module

**Input:** $f_n$ as the feature outputs of c3 module in CSPDarknet53

**Output:** $\mathcal{L}_{mix}$

$f_n = Conv(f_n)$ to fully extract features of each scale
$\mathcal{F} = f_0$
**for** $f_i$ in $f_1 \rightarrow f_n$ **do**
  $\mathcal{F} = pooling(\mathcal{F})$ to become the same size with $f_{i+1}$
  $\mathcal{F} = \mathcal{F} \oplus f_{i+1}$ in equal proportions of the elements
**end for**
build domain classes ground Truth label $Y^m$ according to $X^m$
$\mathcal{L}_{mix} = DomainClassifier\ (\mathcal{F}, Y^m)$

The overall calculation process of UAM represented in Fig. 4 can be simplified as follows:

$$p_i = E(D(X_i)) \tag{6}$$

where the feature encoder, denoted as $E(\cdot)$, represents the multiscale unbiased feature extraction layers in UAM, while $D(\cdot)$ represents the domain classifier network in UAM and endeavors to differentiate the source domain from the target domain based on the derived features from $E(\cdot)$. The idea of the adversarial optimization objective function in UAM is shown as follows:

$$\mathcal{L}_{\text{adv}} = \max_E \min_D \mathcal{L}_{\text{dis}} \tag{7}$$

where $\mathcal{L}_{\text{dis}}$ is the loss calculate by domain classifier, $\mathcal{L}_{\text{adv}}$ is the loss we hope to backpropagate. The equation means that we want the domain classifier can distinguish the source and target domains well and get the minimum classification loss $\mathcal{L}_{\text{dis}}$, but we want the feature encoder to maximize the classification loss to obtain the network weights with the extracted domain-invariant features. To generate domain-invariant features,

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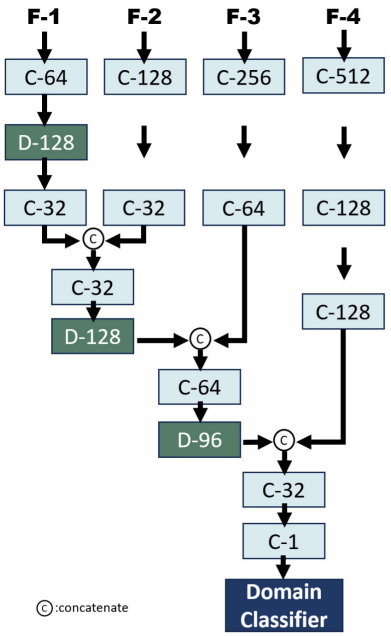



Fig. 4. Specific details of the UAM; the UAM accepts four different scales of feature maps from the CSPDarknet53 network as inputs to extract domain-invariant features. F-$k$ represents the input features at four scales. C-$i$ represents the convolution operation without a down sample, where $i$ represents the feature dimension of the output of the convolution operation. D-$j$ represents the convolution operation with a single-fold down sample, and $j$ represents the feature dimension of the output of the convolution operation.

the final model should consistently maximize $\mathcal{L}_{\text{dis}}$. This ensures that the extracted features cannot be classified separately by the classifier. In UAM, we added gradient reverse layers [28] for each scale of the income feature to eliminate the need for the computation of $\mathcal{L}_{\text{adv}}$, and only $\mathcal{L}_{\text{dis}}$ is required.

Since the mixed images contain different domain images and information, it is not possible to design the domain classifier loss directly using image classification methods. Therefore, we developed a pixel-level domain classifier to tackle this problem. The final output of the feature encoder is an $N \times N$ matrix where each value represents a pixel-level domain prediction. The final loss is obtained by accumulating and averaging the loss functions derived for each value. The loss function for the domain classifier can be expressed as follows:

$$\mathcal{L}_{\text{dis}} = \frac{1}{N \times N} \sum_{x,y} \left[ t_i^{(x,y)} \ln p_i^{(x,y)} + \left(1 - t_i^{(x,y)}\right) \ln\left(1 - p_i^{(x,y)}\right) \right] \tag{8}$$

where $t_i^{(x,y)}$ represents the domain label of the training image in the location of $(x, y)$ and $p_i^{(x,y)}$ denotes the prediction of the UAM. As for the source-domain part of mixed images, $t_i^{(x,y)}$ is 1 and for the target-domain part is 0.

The overall loss function of the CR-Mixing framework is written as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \gamma \mathcal{L}_{\text{mix}} + \lambda \mathcal{L}_{\text{dis}} \tag{9}$$

where $\mathcal{L}_{\text{sup}}$ represents the supervised detection loss of student model, $\gamma$ is a learnable parameter for controlling the weights of $\mathcal{L}_{\text{mix}}$ according to the accuracy of detections, and $\lambda$ is a hyperparameter to balance the weight of $\mathcal{L}_{\text{dis}}$ in total loss $\mathcal{L}_{\text{total}}$. The definition of $\gamma$ is given as follows [57]:

$$\gamma = \frac{|\{\tilde{y}_m^i \in \tilde{\mathbf{y}}_m^i : C^i \geq C_{th}^{\gamma}\}|}{|\tilde{\mathbf{y}}_m^i|} \tag{10}$$

where $\tilde{y}_m^i$ is the prediction of the student model using mix domain data $X_m$ and $\tilde{\mathbf{y}}_m^i$ is the set of all predictions after nonmaximum suppression, including the predicted values in the source, target, and mix domains. $C^i$ is the confidence of $\tilde{y}_m^i$, and $C_{th}^{\gamma}$ is the setting threshold to select the prediction, which is reliability. $|\cdot|$ is the cardinality of a set.

## IV. Experiment

We conducted the experiment on an RTX 3090 GPU. Our baseline object detector is YOLOv5. Following [43], we developed two optical remote sensing domain adaptation object detection benchmarks using the NWPU VHR-10 [58] dataset as the source domain and DIOR [20] and HRRSD [59] datasets as the target domains. These benchmarks were used to evaluate our proposed method against comparison methods, along with conducting comprehensive ablation studies to demonstrate the effectiveness of our model.

### A. Dataset

*1) NWPU VHR-10 [58]:* This dataset comprises 800 images, including 715 high-spatial-resolution RGB images sourced from Google Earth and 85 very high-spatial-resolution pan-sharpened color infrared images (CIR). In the NWPU VHR-10 dataset, the spatial resolution of Google Earth images ranges from 0.5 to 2 m, while the CIR images have a resolution of 0.08 m. Among the 800 images, 650 have been selected in our experiment to form the source-domain dataset, encompassing ten categories of objects. When conducting the experiment, since the number of source-domain data is not equal to the other two datasets, we implemented data augmentation to expand the NWPU VHR-10 dataset. In this way, we can repeatedly utilize the data from the source domain to learn various representations, which can greatly aid in the final alignment process.

*2) DIOR* [20]:* This dataset comprises 23 463 optical remote sensing RGB images, including 192 472 objects distributed across 20 common categories. All images within the dataset are sourced from Google Earth, with a spatial resolution ranging from 0.5 to 30 m. Notably, in contrast to the NWPU VHR-10 dataset, the DIOR dataset presents a wider range of image variations. It covers remote sensing images from over 80 countries, leading to more diverse differences within each class. These differences include variations in perspective, obstruction, visual representation, backdrop, object orientation, and other factors. To facilitate domain adaptation object detection tasks, we curated the DIOR* dataset by selecting the same ten categories as those in the NWPU VHR-10 dataset. The DIOR* dataset comprises a total of 5862 images and 28 527 objects.

*3) HRRSD* [59]:* The HRRSD dataset comprises 13 categories, totaling 26 722 optical remote sensing images.

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TABLE I
DETAIL OF LABELS

| Dataset | NWPU VHR-10 | DIOR* | HRRSD* |
|---|---|---|---|
| Airplane | 757 | 1238 | 2446 |
| Ship | 302 | 12 994 | 2007 |
| Storage tank | 655 | 1344 | 2277 |
| Baseball field | 390 | 1244 | 2032 |
| Tennis court | 524 | 561 | 2168 |
| Basketball court | 159 | 2240 | 2084 |
| Ground track field | 163 | 565 | 2014 |
| Harbor | 224 | 1144 | 1868 |
| Bridge | 124 | 508 | 2305 |
| Vehicle | 598 | 6689 | 2392 |
| Image number | 650 | 5862 | 10 880 |
| Instance number | 3896 | 28 527 | 21 593 |

Among these, 21 761 images exhibit a spatial resolution ranging from 0.15 to 2 m and are sourced from Google Earth, while additional 4961 images, characterized by a spatial resolution of 0.6 to 2 m, are acquired from Baidu Maps. In addition, we extracted the same ten categories from the NWPU VHR-10 dataset and removed the images without objects, creating a subset referred to as HRRSD*. This curated subset comprises a total of 10 880 images and 21 593 objects. For a comprehensive breakdown of object distribution across each category, Table I enables a comparative analysis among the NWPU VHR-10, DIOR*, and HRRSD* datasets.

### B. Implementation Details

We utilized the YOLOv5l architecture with CSPDarknet53 as the backbone feature extraction network to conduct the experiments. We set the batch size to 2, with each batch containing a source image and a target image sized at 640 × 640. For dataset partitioning, all the images in the NWPU VHR-10 dataset are selected for the training set. As for the target-domain datasets, the ratio of validation to testing data is 6:4.

In all of our experiments, we initialized the model with coco-pretrained model weights and conducted pretraining on the source domain for 20 epochs. Subsequently, the UDA object detection task is carried out for an additional 100 epochs.

We utilize the average precision (AP) as a metric to assess the performance of the CR-Mixing framework. AP is calculated based on precision and recall, which are defined as

$$\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{11}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{12}$$

where $N_{\text{TP}}$, $N_{\text{FP}}$, and $N_{\text{FN}}$ represent the number of true positives (TPs), false positives (FPs), and false negatives (FNs) respectively. TP denotes that the object is accurately identified, FP represents the object that is mistakenly detected, and FN indicates the instances of objects missed in the detection results. The specific equations are given as follows:

$$\text{AP} = \frac{1}{n} \sum_{r \in S} \text{Precision}|_{\text{Recall}=r} \tag{13}$$

$$\text{mAP} = \frac{1}{c} \sum_{i=0}^{c} \text{AP}_i \tag{14}$$

where AP is subsequently computed as the mean precision across a range of equally spaced recall rates denoted by set $S$. We selected $S = \{0, 0.1, \ldots, 1\}$, comprising a total of 11 numbers. mAP is a comprehensive evaluation metric calculated by averaging the mean value of the AP for each class. $c$ represents the classes of the dataset. mAP is commonly utilized to assess and compare the performance of different models. During the calculation of AP, a predicted outcome is labeled as a TP if its intersection over union (IoU) with the ground-truth labels is greater than 0.5; otherwise, it is categorized as an FP. In addition, in scenarios where multiple objects have a high IoU with the ground truth, the object with the highest detection confidence is usually classified as a TP.

### C. Experimental Results and Analyzes

We compare the CR-Mixing method with recent domain adaptation object detectors using two different approaches built on three remote sensing object detection benchmarks. We incorporate the *Source Only*, which represents the result using a model trained exclusively with labeled source-domain data to predict the target-domain data. In addition, we also incorporate *Oracle*, denoting the detector model trained with labeled target data and predicting fully target-domain data, acting as the best performance benchmark. The specific experimental settings and results are presented as follows.

*1) NWPU VHR-10 to DIOR*:* In this section, we establish the domain adaptation object detection task using the NWPU VHR-10 dataset as the source domain and the DIOR* dataset as the target domain. We test the models using the validation set to assess the effectiveness of the domain adaptation methods. The results of the experiment are shown in Table II.

The CR-Mixing model achieves 54.3 mAP, making it the best-performing method among the listed cross-domain object detection methods. This result is 28.4 higher than the *Source Only* mAP result. SWDA, HTCN, and SCL were initially proposed for computer vision tasks, so they may not perform better in remote sensing scenarios. In addition, they chose the outdated Faster R-CNN detector to conduct the domain adaptation object detection task. RFA-Net previously demonstrated relatively good performance in optical remote sensing domain adaptation object detection tasks. However, it struggled to effectively address the issue of model bias toward source-domain information features. AT is the method that uses the mean-teacher framework to achieve detection. It also includes a module for domain-invariant feature learning, but it is limited by a single-scale network structure and cannot extract domain-invariant features deeply. The ConfMix method proposes a technique to mix source- and target-domain images based on confidence information. However, it does not utilize the mixed images to constrain the model and learn the target-domain features. SSDA-YOLO generates auxiliary data by combining the GAN method and utilizes the mean-teacher framework to enhance detection stability, but the process is explicit and does not support end-to-end applications. The visualizations of detections can be seen in Fig. 5.

*2) NWPU VHR-10 to HRRSD*:* In this part, we utilize the NWPU VHR-10 dataset as the source domain and the

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TABLE II

MEASURABLE COMPARISON WITH THE SOURCE-ONLY, ORACLE, AND OTHER CROSS-DOMAIN OBJECT DETECTION APPROACHES IN THE EXPERIMENTS CARRIED OUT ON THE NWPU VHR-10 TO DIOR* BENCHMARK

| Methods | Detector | Backbone | Airplane | Ship | Storage tank | Baseball field | Tennis court | Basketball court | Ground track field | Harbor | Bridge | Vehicle | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Source only | YOLOv5 | CSP-Darknet53 | 44.8 | 1.1 | 19.1 | 64.4 | 68.4 | 13.1 | 14.1 | 0.6 | 2.1 | 31.8 | 25.9 |
| SWDA [38] | Faster R-CNN | ResNet101 | 54.9 | 10.4 | 43.1 | 84.7 | 83.2 | 58.1 | 56.5 | 20.4 | 8.9 | 25.7 | 44.6 |
| HTCN [41] | Faster R-CNN | ResNet101 | 58.6 | 13.3 | 44.7 | 85.3 | 82.4 | 53.3 | 51.6 | 22.6 | 9.9 | 21.3 | 44.3 |
| SCL [40] | Faster R-CNN | ResNet101 | 69.3 | 16.7 | 45.2 | 85.7 | 86.0 | 50.7 | 64.1 | 22.6 | 7.3 | 29.2 | 47.7 |
| RFA-Net [43] | Faster R-CNN | ResNet101 | 77.6 | 13.7 | 48.3 | **88.7** | 89.7 | 66.7 | 69.0 | 20.1 | 13.5 | 30.5 | 51.8 |
| AT [46] | Faster R-CNN | ResNet101 | 60.7 | 7.5 | 42.7 | 83.6 | 86.9 | 71.7 | 54.6 | 21.7 | 7.5 | 41.5 | 47.8 |
| SIGMA [60] | FCOS | VGG16 | 46.2 | 15.5 | 47.2 | 80.1 | 83.7 | 67.0 | 38.5 | 11.8 | 5.5 | 38.8 | 43.4 |
| HT [47] | FCOS | VGG16 | 60.6 | 7.0 | **50.8** | 83.5 | 85.8 | 67.1 | 47.6 | 19.5 | 6.0 | 39.0 | 46.7 |
| ConfMix [57] | YOLOv5 | CSP-Darknet53 | 47.4 | 9.2 | 37.4 | 78.3 | 84.4 | 60.9 | 36.7 | 16.9 | 4.5 | 38.2 | 41.4 |
| SSDA-YOLO [61] | YOLOv5 | CSP-Darknet53 | **81.9** | **18.2** | 32.7 | 72.6 | 82.8 | **81.1** | 59.3 | 12.1 | **13.9** | 16.5 | 47.1 |
| CR-Mixing(ours) | YOLOv5 | CSP-Darknet53 | 75.9 | 15.7 | 50.4 | 85.2 | **91.1** | 65.7 | **76.5** | **26.0** | 12.9 | **43.8** | **54.3** |
| Oracle | YOLOv5 | CSP-Darknet53 | 95.4 | 94.9 | 89.8 | 95.7 | 93.4 | 87.1 | 86.4 | 72.3 | 60.2 | 84.6 | 86.0 |

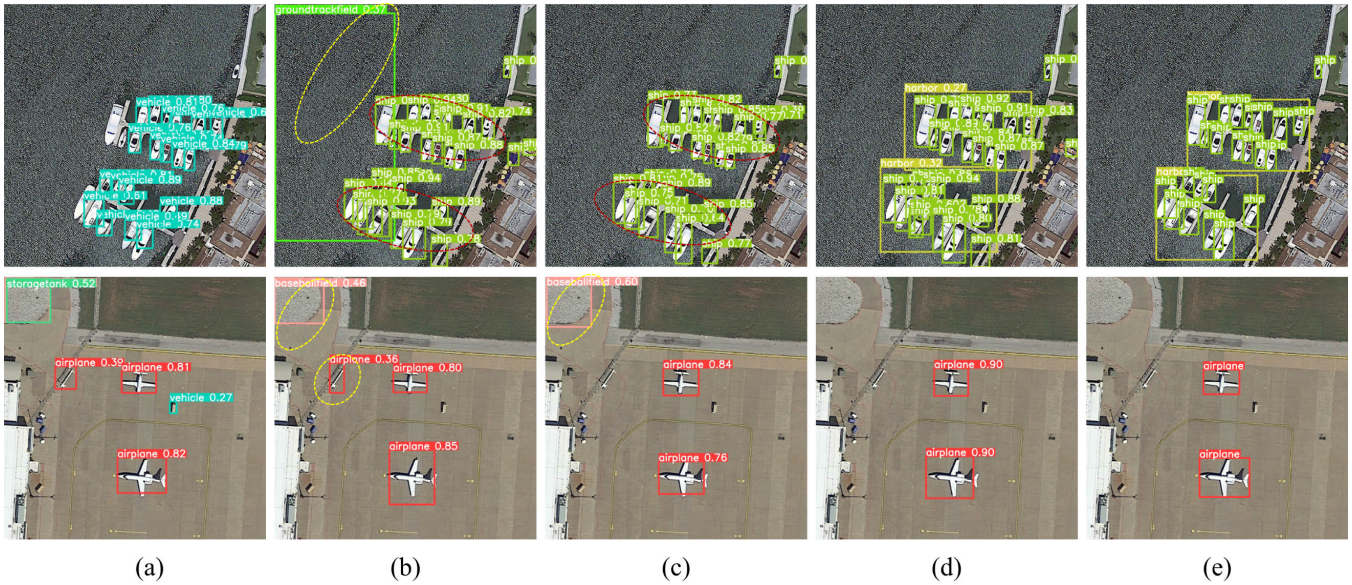



Fig. 5. Performance of various models. (a) *Source Only* results. (b) Detection results of the SSDA-YOLO model [61]. (c) Detection results of the ConfMix model [57]. (d) Detection results of our proposed model. (e) Ground-truth labels. The yellow circles indicate incorrect detections, while the red circles indicate missing objects. We set the threshold of confidence for a positive detection to 0.25 and the threshold of IoU to 0.5.

HRRSD* dataset as the target domain for domain adaptation object detection. Model evaluations are conducted on the validation set to assess the effectiveness of domain adaptation methods. The experimental results are presented in Table III.

The CR-Mixing model emerges as the most promising among the listed domain adaptation object detection methods, achieving an mAP of 59.2. This performance surpasses the *Source Only* mAP result by 27.1. Compared to the DIOR* dataset, the HRRSD* dataset has a similar object distribution to the NWPU VHR-10 dataset, resulting in better performance than DIOR*. Fig. 6 shows the visualizations of detections.

### D. Ablation Study

This section first studies the pretrained model to make informed selections. Following that, a comprehensive analysis of the parameters is conducted to identify the most suitable hyperparameters. In addition, we conducted an ablation study to clarify the effectiveness of each module by demonstrating their impact on the overall performance. We utilize the benchmark of NWPU VHR-10 to DIOR* for conducting the ablation study.

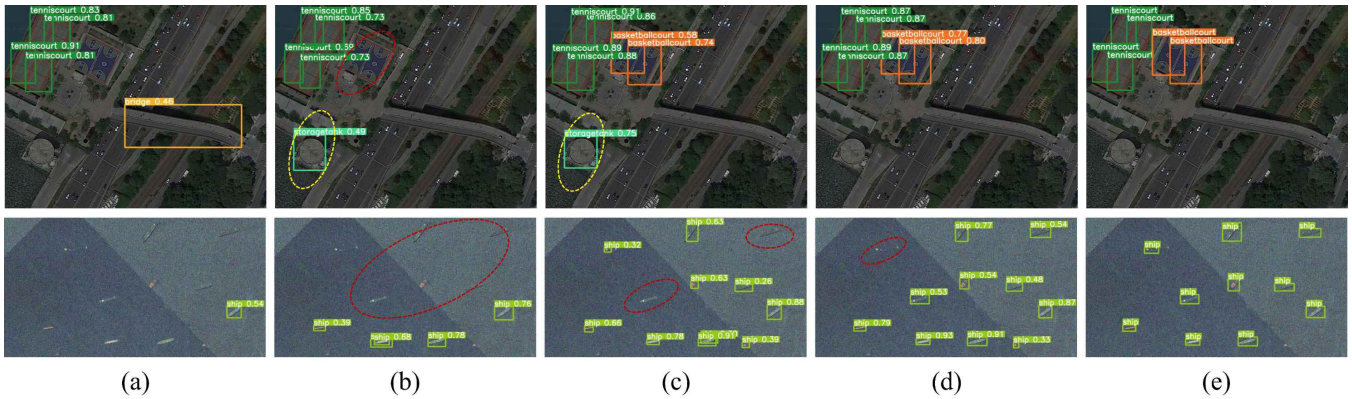
*1) Effectiveness of Different Baseline Pretraining Models:* To initiate our experimentation, the selection of a suitable base pretrained model is crucial. The YOLOv5 model provides five distinct pretrained variants: YOLOv5n, YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x, each differing in terms of parameter count and the number of convolutional layers. YOLOv5n represents the smallest model with the fewest parameters, floating-point operations (FLOPs), and the fastest computing speed (frames per second, FPS), while YOLOv5x is the largest and the slowest.

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TABLE III

MEASURABLE COMPARISON WITH THE SOURCE-ONLY, ORACLE, AND OTHER CROSS-DOMAIN OBJECT DETECTION APPROACHES IN THE EXPERIMENTS CARRIED OUT ON THE NWPU VHR-10 TO HRRSD* BENCHMARK

| Methods | Detector | Backbone | Airplane | Ship | Storage tank | Baseball field | Tennis court | Basketball court | Ground track field | Harbor | Bridge | Vehicle | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Source only | YOLOv5 | CSP-Darknet53 | 77.6 | 27.3 | 27.6 | 18.0 | 49.1 | 12.5 | 40.7 | 22.0 | 7.6 | 38.5 | 32.1 |
| SWDA [38] | Faster R-CNN | ResNet101 | 71.3 | 23.1 | 35.9 | 26.1 | 57.0 | 30.9 | 37.7 | 46.1 | 46.2 | 52.3 | 42.7 |
| HTCN [41] | Faster R-CNN | ResNet101 | 72.2 | 28.4 | 54.2 | 19.8 | 63.5 | 26.1 | 52.9 | 42.6 | 41.5 | 61.3 | 46.3 |
| SCL [40] | Faster R-CNN | ResNet101 | 71.8 | 26.5 | 54.3 | 21.7 | 64.1 | 29.6 | 54.8 | 46.2 | 44.6 | 53.7 | 46.7 |
| RFA-Net [43] | Faster R-CNN | ResNet101 | 85.6 | 41.6 | **63.7** | 22.5 | 68.1 | 25.6 | **76.9** | **58.3** | 45.6 | 64.9 | 55.3 |
| AT [46] | Faster R-CNN | ResNet101 | **93.0** | 34.6 | 59.7 | 13.3 | 74.4 | 29.9 | 58.8 | 54.6 | **53.7** | **78.1** | 55.0 |
| SIGMA [60] | FCOS | VGG16 | 87.9 | 34.9 | 34.8 | 15.9 | 66.0 | 15.2 | 56.9 | 34.8 | 28.8 | 54.8 | 43.0 |
| HT [47] | FCOS | VGG16 | 57.5 | 8.5 | 48.4 | 44.3 | 86.5 | 42.9 | 47.0 | 33.0 | 18.1 | 40.0 | 42.6 |
| ConfMix [57] | YOLOv5 | CSP-Darknet53 | 52.2 | 8.2 | 26.7 | **61.8** | **90.6** | 64.5 | 46.3 | 30.5 | 9.4 | 45.4 | 43.6 |
| SSDA-YOLO [61] | YOLOv5 | CSP-Darknet53 | 86.7 | 34.6 | 47.6 | 14.3 | 69.7 | 22.9 | 55.1 | 52.1 | 35.2 | 71.5 | 49.0 |
| CR-Mixing(ours) | YOLOv5 | CSP-Darknet53 | 91.3 | **44.3** | 43.5 | 58.3 | 84.2 | **68.2** | 55.5 | 35.4 | 48.2 | 63.5 | **59.2** |
| Oracle | YOLOv5 | CSP-Darknet53 | 99.3 | 96.3 | 97.7 | 94.1 | 97.7 | 76.7 | 98.2 | 96.9 | 92.9 | 97.9 | 95.8 |



(a) (b) (c) (d) (e)

Fig. 6. Performance of various models. (a) *Source Only* results. (b) Detection results of the SSDA-YOLO model. (c) Detection results of the ConfMix model. (d) Detection results of our proposed model. (e) Ground-truth labels. The yellow circles indicate incorrect detections, while the red circles indicate missing objects. We set the confidence threshold for positive detection to 0.25 and the IoU threshold to 0.5.

TABLE IV

ABLATION STUDY OF THE BASELINE PRETRAINED MODEL TO SELECT FOR OUR PROPOSED CR-MIXING MODEL

| model | params(M) | FLOPs (B) | mAP | FPS |
|---|---|---|---|---|
| YOLOv5n | 1.9 | 4.5 | 49.4 | 56.5 |
| YOLOv5s | 7.2 | 16.5 | 51.3 | 37.7 |
| YOLOv5m | 21.2 | 49 | 51.7 | 32.2 |
| YOLOv5l | 46.5 | 109.1 | **54.3** | 25.9 |
| YOLOv5x | 86.7 | 205.7 | 53.7 | 20.9 |

In our experiment, all five models were applied to the proposed method to evaluate their contributions and determine the optimal pretrained model. The results, presented in Table IV, reveal that YOLOv5l, the second-largest model with 46.5 million parameters and 109.1 billion FLOPs, emerges as the most effective. Consequently, for subsequent ablation studies, we exclusively employ the YOLOv5l model as our baseline pretrained model.

*2) Effectiveness of Different Mixing Strategies:* In this part, we discuss the effectiveness of different categories of mixing methods and demonstrate it through extensive experiments, as shown in Fig. 7. CutMix [56] is a classical image enhancement strategy that is simple and effective in supervised learning by randomly cropping a region in an image and mixing two images to get an image containing different domains. The remaining four mixing methods are different kinds of CR-Mixing methods based on confidence, including those based on horizontal slicing mixing, based on vertical slicing mixing, based on the ConfMix [57] method that is a four-division mixing method, and the final RSM strategy in CR-Mixing. The results of different strategies can be seen in Table V.

Our proposed CR-Mixing method achieves optimal results, which takes into account the importance of foreground information and mixes the most accurate part of the foreground information of the target-domain images with the labeled source-domain images according to the different training phases and the attribute features of the various images, which makes the training phase more stable and at the same time improves the accuracy. We also constructed a ConfMix-based mixing approach, but we just changed the proposed RSM method to the ConfMix method, keeping the mean teacher model and the UAM module. The final experimental results

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TABLE V
RESULTS OF MODEL PERFORMANCE UNDER DIFFERENT MIXING STRATEGIES

| Method | mAP |
|---|---|
| CutMix | 45.7 |
| CR-Mixing(Vertical Mix) | 42.6 |
| CR-Mixing(Horizontal Mix) | 44.8 |
| CR-Mixing(ConfMix) [61] | 47.3 |
| CR-Mixing | **54.3** |

TABLE VI
PERFORMANCE OF OUR PROPOSED METHOD IN NWPU VHR-10 TO DIOR* WITH DIFFERENT THRESHOLDS OF $T_m$ AND $T_v$ VALUES

| $T_m$ \ $T_v$ | 0.1 | 0.2 | 0.3 | 0.4 |
|---|---|---|---|---|
| 0.4 | 52.9 | 53.3 | 47.0 | 51.7 |
| 0.5 | 49.0 | 51.1 | 48.3 | 50.9 |
| 0.6 | 49.7 | **54.3** | 53.5 | 50.7 |
| 0.7 | 53.5 | 52.7 | 52.9 | 49.1 |

show that the results of the method retaining our proposed modules are superior to ConfMix, again proving the effectiveness of our proposed method.

*3) Effectiveness of Mean and Variance Thresholds $T_m$, $T_v$:* In this part, we conduct a thorough analysis of the two hyperparameters in the RSM module to determine the optimal parameter selection. First, choosing a larger mean threshold $T_m$ value leads to a trend toward having a larger share of the source-domain data in the mixing process. This is because at the beginning, the model is unfamiliar with the knowledge of the target-domain data, and the confidence in the target-domain data is generally low. Therefore, at this point, the confidence is lower, and the mixing approach is more likely to favor a high source-domain share. Conversely, a low confidence implies a bias toward a high target-domain share. It should be noted that the image mixing process should be based on the detection results of the model in the target domain, rather than being fixed.

For the variance value, a smaller variance represents less fluctuation in the data and less difference between individual prediction and the overall mean values and is more representative of the whole. Therefore, we prefer to choose a smaller variance as the threshold. However, if the selected threshold is too small, it will produce few scenarios that meet the standard, resulting in the occurrence of mixing failure results. Therefore, we obtained the most appropriate results for the mean and variance by conducting numerous experiments.

It has been demonstrated through numerous experiments that the proposed model achieves the best experimental results when the mean threshold $T_m$ is 0.5 and the variance threshold $T_v$ is 0.2, as shown in Table VI. These thresholds were obtained through our experiments in the context of the domain adaptation object detection task for the benchmark of NWPU VHR-10 to DIOR*; it is also sufficiently representative to be used as a parameter in a broader range of practical applications without the necessity of conducting numerous experiments each time.

*4) Effectiveness of Hyperparameter $\lambda$:* The hyperparameter $\lambda$ is used to regulate the weights of individual loss functions

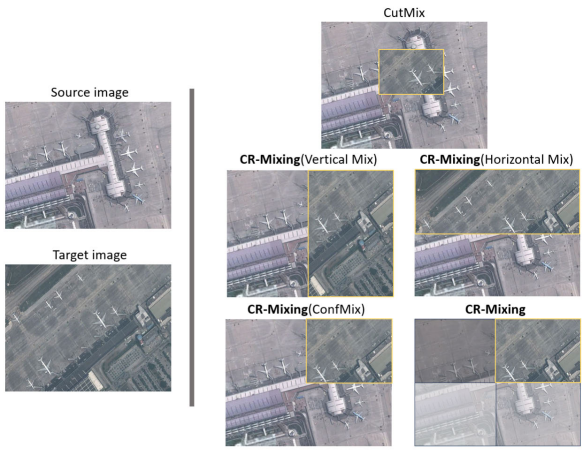



Fig. 7. Illustration of various mixing strategies. On the left are examples of selected source- and target-domain images. On the right are five different mixing methods. CutMix [56] is an image enhancement method that randomly crops parts of an image and stitches them together with another image. Vertical mix and horizontal mix are also confidence-based mixing methods that perform different mixing based on the confidence information of the target-domain data. ConfMix [57] is a type of mixing in which the image is divided into four equal parts and the part with the highest confidence is selected to mix. The final CR-Mixing method is our proposed mixing strategy.

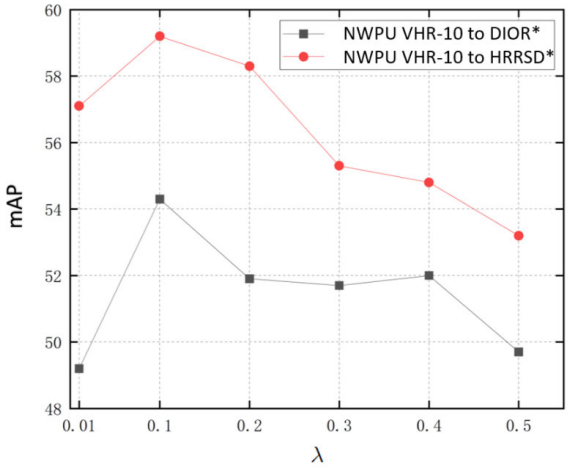



Fig. 8. Specific performance of the model at different $\lambda$ values.

in the model. Its primary function is to ensure that the weights of each loss function remain approximately equal, which helps to smooth out differences in numerical magnitude between the different loss functions. This is important because when the loss function values have large differences, the component parameters update slower when using back propagation. Therefore, by utilizing this hyperparameter to maintain relatively equal weights, we can prevent slowing down the update of component parameters during backward propagation.

The result can be seen in Fig. 8. When $\lambda$ is set as 0.1, the final result of the model achieves the best performance of overall mAP compared to other different values.

*5) Effectiveness of Each Component:* To assess the effectiveness of individual components within CR-Mixing, we conduct ablation experiments by isolating each module (reprocessing enhancement (RE), RSM strategy, and UAM) and examining their impact. In addition, we conduct experi-

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TABLE VII
ABLATION STUDY OF THE MODULE OF OUR PROPOSED CR-MIXING MODEL

| Method | RE | RSM | UAM | mAP |
|---|---|---|---|---|
| Source Only | | | | 26.5 |
| MT | | | | 34.7 |
| MT w/ module | ✓ | | | 35.2 |
| | | ✓ | | 42.6 |
| | | | ✓ | 47.1 |
| | ✓ | ✓ | | 46.0 |
| | ✓ | | ✓ | 43.9 |
| | | ✓ | ✓ | 52.7 |
| CR-Mixing(ours) | ✓ | ✓ | ✓ | **54.3** |
| Oracle | | | | 86.0 |

ments in which each of these modules is completely removed from the final framework. The outcomes of these experiments are presented in Table VII. It is crucial to note that in the ablation experiment involving the mean-teacher (MT) framework adopted by our method, we compared the results by conducting ablation experiments on a simple MT structure without any additional modules. We can see that when the model consists only of the mean-teacher network, its results are improved compared to the results of the *Source Only* model, but the improvement is not significant; it is about 4.2 of mAP. The mean-teacher framework does not address the issue of domain-shift. As the model incrementally incorporates components and methods, the experimental results consistently show improvement. Using the reprocessing enhancement method for mixed images to increase sample diversity, the results improved by 0.5 in terms of mAP. The RSM module showed a 7.1 improvement, while the UAM demonstrated a 12.4 improvement, highlighting the significance of UAM in enhancing results.

For the missing part of modules, the result in the situation without reprocessing enhancement, RSM, and UAM is 1.6, 10.4, and 8.3 lower than the optimal result, respectively. It should be noted that due to the use of an adversarial network, the prediction results of the model fluctuate. Therefore, the final outcome of each experiment is not consistently the same, and our result represents the optimal outcome from multiple experiments.

To demonstrate the effectiveness of UAM, we visualize the feature map output from the first three layers of the CSPDarknet53 in Fig. 9. It is evident that the backbone feature extraction network of the scripted UAM module emphasizes objects to deceive the domain classifiers into assigning greater weights to foreground regions. This illustrates the capability to acquire domain-independent object representations. We also created a feature attention diagram for the domain adaptation detection task to illustrate the overall focus of various models when different modules are utilized, as shown in Fig. 10. The model cannot fully focus on the necessary objects in the baseline framework. When the UAM was incorporated, the model exhibited improved performance. Subsequently, the addition of the RSM completed the CR-Mixing method, resulting in the best focus. All the desired objects in the graph were in focus and received higher weights.

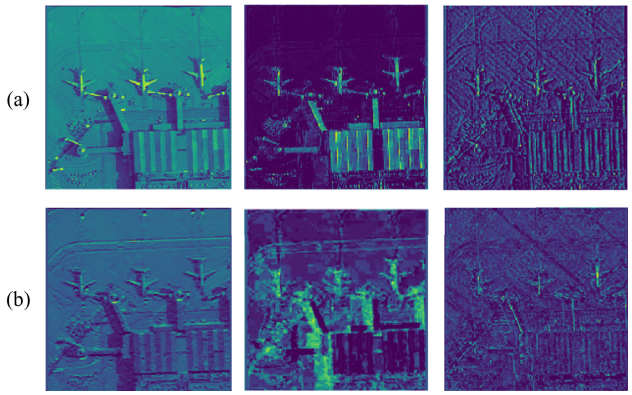



Fig. 9. Evidence that shows the UAM works. The three columns of images are, from left to right, the output of the first two Conv layers and the first C3 module of the CSPDarknet. (a) Outputs without UAM. (b) Outputs with UAM but without RSM.

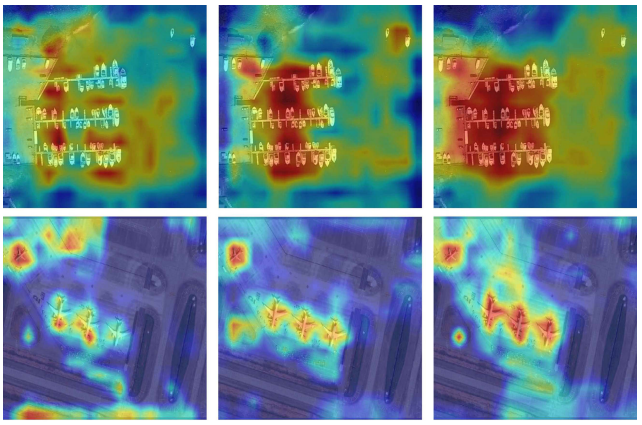



Fig. 10. Attention map of different strategies. (a) Baseline of CR-Mixing without RSM and UAM is just a mean-teacher framework. (b) Baseline with UAM. (c) CR-Mixing framework. The weights increase from blue to red, and the higher the weight, the more helpful it is for detection. Adding UAM and RSM accordingly makes the model better for detection.

These experiments validate the efficacy of the three modules, demonstrating their high degree of complementarity. The strong interconnection and integration of these components within a unified network during training further underscore their collaborativeness.

## V. CONCLUSION

In this article, we introduce the CR-Mixing model, a novel framework designed for UDA in remote sensing object detection tasks. The primary objective of the CR-Mixing model is to alleviate the domain-shift present across diverse RS imagery domains and enable the transfer of knowledge from a limited dataset to a large-scale dataset. The architecture consists of three fundamental modules: the mean-teacher framework, RSM strategy, and UAM. These modules are meticulously designed, demonstrating coherence and complementarity, ultimately contributing to improved domain-adaptive results. Addressing the limitation of the mean-teacher framework, which does not explicitly tackle the challenge of domain-shift, we leverage the region-level mixed image generated by the RSM module as a substitute for the traditional target-domain

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image. This substitution aims to enhance foreground emphasis, ensuring maximal information inclusion and the generation of highly confident pseudodetections. Simultaneously, to foster a balanced focus on both domains, we introduce the UAM equipped with a multiscale feature extractor. This is to counteract model bias toward the source domain, yielding more resilient, domain-invariant features that mitigate the impact of domain-shift. The experimental results demonstrate the effectiveness of our proposed methodology in enhancing the generalization capabilities of object detection models applied to remote sensing imagery, with a focus on robustness and adaptability. In the future, we will focus more on effectively reducing the domain gap and enhancing the representation of domain-invariant features for optical remote sensing domain adaptation object detection tasks.

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